

Exercise JA

Q1. For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then

- (A) $a + b - c > 0$ (B) $a - b + c < 0$ (C) $a - b + c > 0$ (D) $a + b - c < 0$

Ans. A, C

Sol. Point of intersection of both lines is

$$\left(-\frac{c}{a+b}, -\frac{c}{a+b} \right)$$

Distance between $\left(-\frac{c}{a+b}, -\frac{c}{a+b} \right)$ & $(1, 1)$ is

$$\text{Distance} = \sqrt{\frac{(a+b+c)^2}{(a+b)^2} \times 2} < 2\sqrt{2}$$

$$a + b + c < 2(a + b)$$

$$a + b - c > 0$$

According to given condition option (C) also correct.

Q2. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \leq d_1(P) + d_2(P) \leq 4$, is

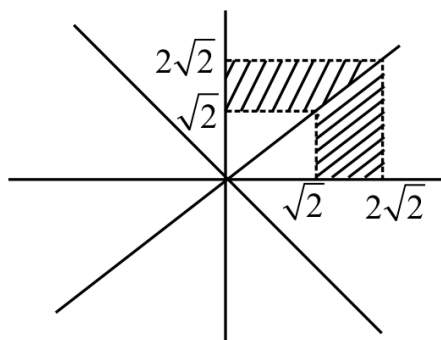
Ans. 6

Sol. let $p(h, k)$

$$2 \leq \left| \frac{h-k}{\sqrt{2}} \right| + \left| \frac{h+k}{\sqrt{2}} \right| \leq 4$$

$$\Rightarrow 2\sqrt{2} \leq |h-k| + |h+k| \leq 4\sqrt{2}$$

If $h \geq k$



$$\Rightarrow 2\sqrt{2} \leq x - y + x + y \leq 4\sqrt{2} \text{ or } \sqrt{2} \leq x \leq 2\sqrt{2}$$

similarly when $h > h$

we have $\sqrt{2} \leq y \leq 2\sqrt{2}$

$$\text{The required area} = (2\sqrt{2})^2 - (\sqrt{2})^2 = 6$$

Q3. Consider the lines L_1 and L_2 defined by

$$L_1 : x\sqrt{2} + y - 1 = 0 \text{ and } L_2 : x\sqrt{2} - y + 1 = 0$$

For a fixed constant λ let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2 . The line $y = 2x + 1$ meets C at two points R and S where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S' . Let D be the square of the distance between R' and S' . The value of λ^2 is _____.

Ans. 9

Sol. Let $P(x, y)$

$$d_1 d_2 = \lambda^2$$

$$\left| \frac{x\sqrt{2} - y + 1}{\sqrt{3}} \right| \left| \frac{x\sqrt{2} + y - 1}{\sqrt{3}} \right| = \lambda^2$$

$$|2x^2 - (y - 1)^2| = 3\lambda^2$$

we will shift origin to $(0, 1)$

$$x - 0 = X ; y - 1 = Y$$

$$|2X^2 - Y^2| = 3\lambda^2 \quad y = 2x + 1 \Rightarrow Y = 2X$$

For R & S

Put $Y = 2X$ in equation of curve C

$$|2X^2 - 4X^2| = 3\lambda^2$$

$$2X^2 = 3\lambda^2$$

$$X = \pm \frac{\sqrt{3}}{\sqrt{2}} \lambda$$

$$R \left(\frac{\sqrt{3}}{\sqrt{2}} \lambda, \frac{2\sqrt{3}}{\sqrt{2}} \lambda \right) \quad S \left(\frac{-\sqrt{3}}{\sqrt{2}} \lambda, \frac{-2\sqrt{3}}{\sqrt{2}} \lambda \right)$$

Midpoint of $RS = (0, 0)$

$$RS \left| \sqrt{5} \times 2 \left(\frac{\sqrt{3}}{\sqrt{2}} \right) \lambda \right| = \sqrt{270}$$

$$30\lambda^2 = 270$$

$$\lambda^2 = 9$$

Q4. Consider the lines L_1 and L_2 defined by

$$L_1 : x\sqrt{2} + y - 1 = 0 \text{ and } L_2 : x\sqrt{2} - y + 1 = 0$$

For a fixed constant λ let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2 . The line $y = 2x + 1$ meets C at two points R and S where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. The value of D is _____.

Ans. 77.14

Sol. Perpendicular bisector of RS

= Line perpendicular to RS & passing through midpoint of RS = (0, 0)

$$X + 2Y = 0$$

for R' and S'

Put $X = -2Y$ in equation of curve C

$$|8Y^2 - Y^2| = 3\lambda^2$$

$$7Y^2 = 3\lambda^2$$

$$7Y^2 = 27$$

$$Y^2 = \frac{27}{7}$$

$$R' \left(-\frac{2\sqrt{27}}{\sqrt{7}}, \sqrt{\frac{27}{7}} \right) \quad S' \left(\frac{2\sqrt{27}}{\sqrt{7}}, \frac{-\sqrt{27}}{\sqrt{7}} \right)$$

$$R'S' = 2\sqrt{5}\sqrt{\frac{27}{7}} = \sqrt{D}$$

$$D = 20 \times \frac{27}{7} = 77.14$$

Exercise C1

Q1. The orthocentre of the triangle ABC is 'B' and the circumcentre is 'S' (a, b). If A is the origin then the co-ordinates of C are :

- (A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) None

Ans. A

Sol. -

Q2. A triangle ABC with vertices A (-1, 0), B (-2, 3/4) & C (-3, -7/6) has its orthocentre H. Then the orthocentre of triangle BCH will be :

- (A) (-3, -2) (B) (1, 3) (C) (-1, 2) (D) none of these

Ans. D

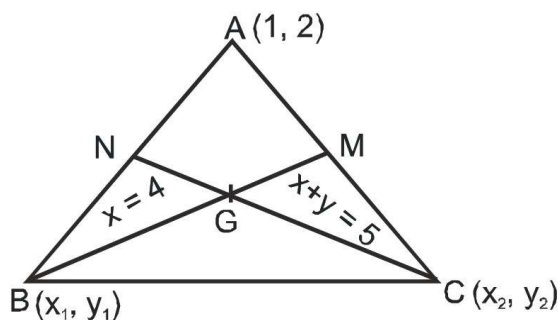
Sol. If H is orthocentre of triangle ABC, then orthocentre of triangle BCH is point A.

Q3. In a triangle ABC, co-ordinates of A are (1, 2) and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the co-ordinates of B and C will be

- (A) (-2, 7), (4, 3) (B) (7, -2), (4, 3) (C) (2, 7), (-4, 3) (D) (2, -7), (3, -4)

Ans. B

Sol.



$$x_1 + y_1 = 5 \quad \dots (i)$$

$$x_2 = 4 \quad \dots (ii)$$

co - ordinates of G are $\equiv (4, 1)$

$$\Rightarrow \frac{1 + x_1 + x_2}{3} = 4 \quad \dots (iii)$$

$$\text{and } \frac{y_1 + y_2 + 2}{3} = 1 \quad \dots (iv)$$

solving above equations, we get B & C.

Q4. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the

ratio

- (A) 1 : 2 (B) 3 : 4 (C) 2 : 1 (D) 4 : 3

Ans. B

Sol. -

Q5. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is

- (A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. B

Sol. -

Q6. The set of values of 'b' for which the origin and the point (1, 1) lie on the same side of the straight line,

$a^2x + a by + 1 = 0 \forall a \in \mathbb{R}, b > 0$ are :

- (A) $b \in (2, 4)$ (B) $b \in (0, 2)$ (C) $b \in [0, 2]$ (D) $(2, \infty)$

Ans. B

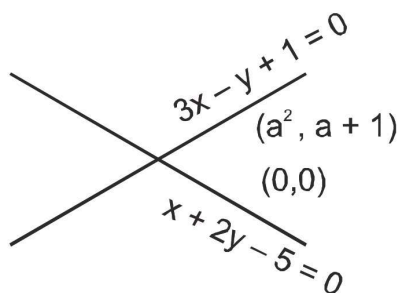
Sol. $a^2x + a by + 1 = 0$ origin and (1, 1) lies on same side. $a^2 + ab + 1 > 0 \quad \forall a \in \mathbb{R}$
 $D < 0 \Rightarrow b^2 - 4 < 0 \Rightarrow b \in (-2, 2) \quad \text{but } b > 0 \Rightarrow b \in (0, 2)$

Q7. The point $(a^2, a + 1)$ is a point in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin if :

- (A) $a \geq 1$ or $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$ (C) $a \in (0, 1)$ (D) none of these

Ans. B

Sol.



Origin & $(a^2, a + 1)$ lies same side w.r.t. to given lines

$$a^2 + 2a + 2 - 5 < 0 \Rightarrow a^2 + 2a - 3 < 0$$

$$\Rightarrow (a + 3)(a - 1) < 0$$

$$\Rightarrow a \in (-3, 1)$$

$$3a^2 - (a + 1) + 1 > 0 \Rightarrow 3a^2 - a > 0$$

$$\Rightarrow a(3a - 1) > 0$$

$$\Rightarrow a \in (\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$$

take intersection we get $a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

Q8. The image of the point A (1, 2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point (α, β) then :

- (A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) none of these

Ans. C

Sol. Image of A(1, 2) in line mirror $y = x$ is (2, 1) ; Image of B(2, 1) in $y = 0$ (x – axes) is 2, – 1)

Hence, $\alpha = 2, \beta = -1$

Q9. The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation

$x - 7y + 5 = 0$. The equation of the other line is :

- (A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) none

Ans. C

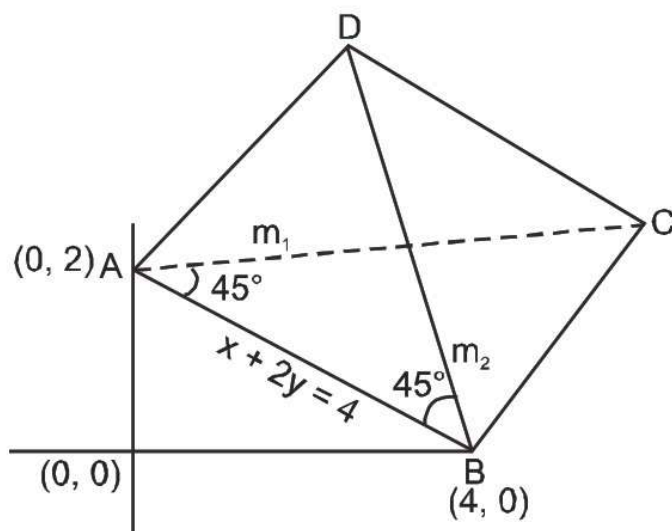
Sol. -

Q10. On the portion of the straight line, $x + 2y = 4$ intercepted between the axes, a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has co-ordinates :

- (A) (2, 3) (B) (3, 2) (C) (3, 3) (D) None

Ans. C

Sol.



$$\tan 45^\circ = \left| \frac{m + \frac{1}{2}}{1 - \frac{m}{2}} \right| \Rightarrow \pm 1 = \frac{2m + 1}{2 - m} \Rightarrow m = \frac{1}{3}, -3$$

∴ Equation of AC

$$y - 2 = \frac{1}{3}(x) \Rightarrow x - 3y + 6 = 0 \quad \dots (i)$$

Equation of BD

$$y = -3(x - 4) \Rightarrow 3x + y - 12 = 0 \quad \dots (ii)$$

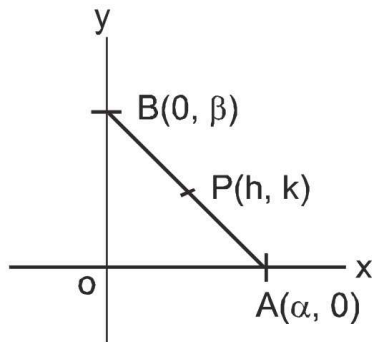
$$(i) \& (ii) \quad x = 3 \& y = 3$$

Q11. AB is a variable line sliding between the coordinate axes in such a way that A lies on x-axis and B lies on y-axis. If P is a variable point on AB such that PA = b, PB = a and AB = a + b, then equation of locus of P is

(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) none of these

Ans. A

Sol.



By geometry

$$\alpha^2 + \beta^2 = (a + b)^2 \quad \dots (i)$$

By section formula

$$h = \frac{\alpha a}{a + b} \Rightarrow \alpha = \frac{h(a + b)}{a}$$

$$k = \frac{\beta b}{a + b} \Rightarrow \beta = \frac{k(a + b)}{b}$$

Put value of a and b in (i)

$$\frac{h^2(a + b)^2}{a^2} + \frac{k^2(a + b)^2}{b^2} = (a + b)^2 \Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = 1$$

$$\text{Locus of P is } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Q12. Equations of the line pair through the origin and perpendicular to the line pair $xy - 3y^2 + y - 2x + 10 = 0$ is :

(A) $xy - 3y^2 = 0$ (B) $xy + 3x^2 = 0$ (C) $xy + 3y^2 = 0$ (D) $x^2 - y^2 = 0$

Ans. B

Sol. Let equations of lines represented by the line pair $xy - 3y^2 + y - 2x + 10 = 0$ are $y + c_1 = 0$, $x - 3y + c_2 = 0$
 lines $\perp$ to these lines and passing through origin are
 $x = 0$, $y = -3x$
 Joint equation
 $x(3x + y) = 0 \Rightarrow xy + 3x^2 = 0$

Q13. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x -axis then the value of k is equal to :

(A) 1 (B) -1 (C) 2 (D) 3

Ans. B

Sol. Homogenize $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ by $x + ky = 1$
 $5x^2 + 12xy - 6y^2 + 4x(x + ky) - 2y(x + ky) + 3(x + ky)^2 = 0$
 it is equally indined with x -axes hence coeff. $xy = 0 \Rightarrow 12 + 4k - 2 + 6k = 0 \Rightarrow k = -1$

Q14. Area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals

(A) $\frac{|m+n|}{(m-n)^2}$ (B) $\frac{2}{|m+n|}$ (C) $\frac{1}{|m+n|}$ (D) $\frac{1}{|m-n|}$

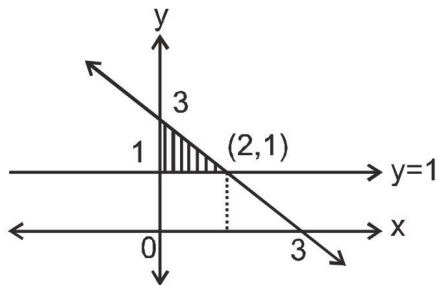
Ans. D

Sol. -

- Q15. The area bounded by the angle bisectors of the lines $x^2 - y^2 + 2y = 1$ and the line $x + y = 3$, is
- (A) 2 (B) 3 (C) 4 (D) 6

Ans. A

Sol.



Bisector of above lines are $x = 0, y = 1$
 so Area between $x = 0, y = 1$ and $x + y = 3$
 $= \frac{1}{2} \times 2 \times 2 = 2$ squ. units.

- Q16. If L is the line whose equation is $ax + by = c$. Let M be the reflection of L through the y -axis, and let N be the reflection of L through the x -axis. Which of the following must be true about M and N for all choices of a, b and c
- (A) The x -intercept of M and N are equal
 (B) The y -intercepts of M and N are equal (C) The slopes of M and N are equal
 (D) The slopes of M and N are reciprocal

Ans. C

Sol. -

- Q17. The complete set of real values of ' a ' such that the point $P(a, \sin a)$ lies inside the triangle formed by the line $x - 2y + 2 = 0$, $x + y = 0$ and $x - y - p = 0$, is:
- (A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{6}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$ (C) $(0, \pi)$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

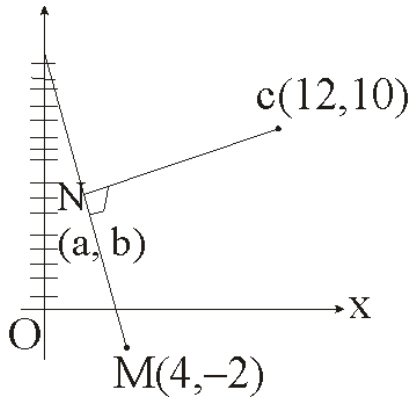
Ans. C

Sol. -

- Q18. A piece of cheese is located at $(12, 10)$ in a coordinate plane. A mouse is at $(4, -2)$ and is running up the line $y = -5x + 18$. at the point (a, b) the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is
- (A) 6 (B) 10 (C) 18 (D) 14

Ans. B

Sol. (a, b) is the foot of the perpendicular from (12, 10) on the line $y + 5x = 18$



$$\Rightarrow \frac{a - 12}{5} = \frac{b - 10}{1} = - \left(\frac{10 + 5(12) - 18}{26} \right) = -2 \Rightarrow (a + b) = (2, 8)$$

Q19. If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to

- (A) $\frac{3}{8}$ (B) $-\frac{3}{4}$ (C) $\frac{1}{4}$ (D) -2

Ans. B

Sol. -

Q20. The graph of the function, $y = \cos x \cos (x + 2) - \cos^2 (x + 1)$ is :

- (A) a straight line passing through $(0, -\sin^2 1)$ with slope 2
 (B) a straight line passing through $(0, 0)$ (C) a parabola with vertex $(1, -\sin^2 1)$
 (D) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to the x-axis.

Ans. D

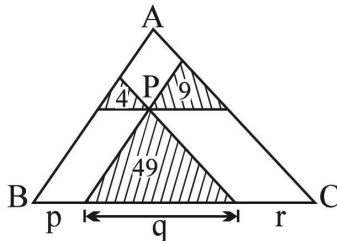
Sol. $2y = \cos(2x + 2) + \cos 2 - (1 + \cos(2x + 2))$
 $= \cos 2 - 1 = -2 \sin^2(1)$
 $y = -\sin^2(1) \Rightarrow (D)$

Q21. P is a point inside the triangle ABC. Lines are drawn through P, parallel to the sides of the triangle. The three resulting triangles with the vertex at P have areas 4, 9 and 49 sq. units. The area of the triangle ABC is

- (A) 121 (B) 100 (C) 81 (D) 144

Ans. D

Sol.



$$\frac{49}{\Delta} = \frac{a^2}{a^2} \quad \text{or} \quad \frac{7}{\sqrt{\Delta}} = \frac{p}{a} \quad \dots(1)$$

$$\frac{2}{\sqrt{\Delta}} = \frac{p}{a} \quad \dots(2)$$

$$\text{and } \frac{3}{\sqrt{\Delta}} = \frac{r}{a} \quad \dots(3)$$

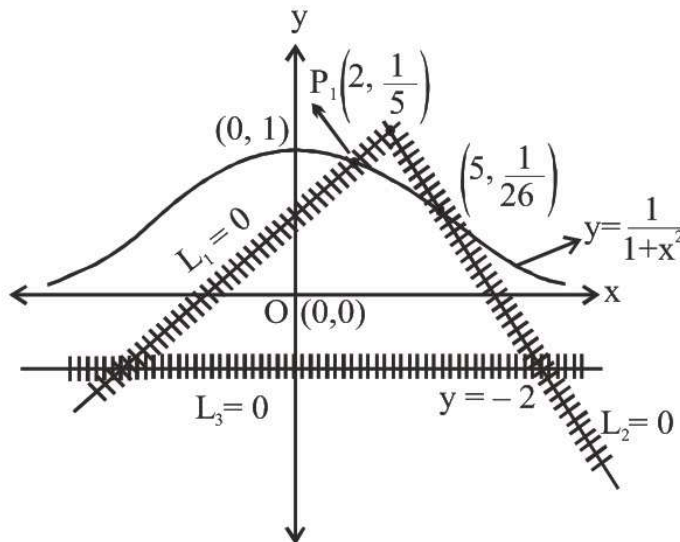
$$(1) + (2) + (3) \text{ gives } \frac{12}{\sqrt{\Delta}} = \frac{p+q+r}{a} = 1 \Rightarrow \Delta = 144$$

Q22. The complete set of values of the parameter a so that the point $\left(\alpha, (1 + \alpha^2)^{-1}\right)$ does not lie outside the triangle formed by the lines $L_1 : 15y = x + 1$, $L_2 : 78y = 118 - 23x$ and $L_3 : y + 2 = 0$, is

- (A) $(0, 5)$ (B) $[2, 5]$ (C) $[1, 5]$ (D) $[0, 2]$

Ans. B

Sol.



As $P(\alpha, (1 + \alpha^2)^{-1})$ lie on $y = \frac{1}{1 + x^2}$ with L_1 .

we get $P_1\left(2, \frac{1}{5}\right) \quad \dots(1)$

and with L_2 , we get $P_2\left(5, \frac{1}{26}\right) \quad \dots(2)$

$\therefore$ From (1) and (2), we get $2 \leq \alpha \leq 5$

Q23. Line $\frac{x}{a} + \frac{y}{b} = 1$ cuts the co-ordinate axes at A(a, 0) and B (0, b) and the line $\frac{x}{a'} + \frac{y}{b'} = -1$ at A' (-a', 0) and B' (0, -b'). If the points A, B, A', B' are concyclic then the orthocentre of the triangle ABA' is :

- (A) (0, 0) (B) (0, b') (C) $\left(0, \frac{aa'}{b}\right)$ (D) $\left(0, \frac{bb'}{a}\right)$

Ans. B, C

Sol. -

Q24. Consider the equation $y - y_1 = m(x - x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1 , then :

- (A) the lines will pass through a fixed point (B) there will be a set of parallel lines
(C) all the lines intersect the line $x = x_1$
(D) all the lines will be parallel to the line $y = x_1$.

Ans. B, C

Sol. -

Q25. If $\frac{x}{c} + \frac{y}{d} = 1$ is a line through the intersection of $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$ and the lengths of the perpendiculars drawn from the origin to these lines are equal in lengths then :

- (A) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2} + \frac{1}{d^2}$ (B) $\frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{c^2} - \frac{1}{d^2}$ (C) $\frac{1}{a} + \frac{1}{b} = \frac{1}{c} + \frac{1}{d}$ (D) None

Ans. A, C

Sol. -

Q26. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2 - 3|x| + 2 = 0$ and the y-coordinates of the vertices are the roots of the equation $y^2 - 3y + 2 = 0$ then the possible vertices of the square is/are :

- (A) (1, 1), (2, 1), (2, 2), (1, 2) (B) (-1, 1), (-2, 1), (-2, 2), (-1, 2)
(C) (2, 1), (1, -1), (1, 2), (2, 2) (D) (-2, 1), (-1, -1), (-1, 2), (-2, 2)

Ans. A, B

Sol. -

Q27. P(x, y) moves such that the area of the triangle formed by P, Q(a, 2a) and R(-a, -2a) is equal to the area of the triangle formed by P, S(a, 2a) and T(2a, 3a). The locus of 'P' is a straight line given by :

- (A) $3x - y = a$ (B) $5x - 3y + a = 0$ (C) $5x - 5y + a = 0$ (D) $2y = ax$

Ans. A, B

Sol. -

Q28. Let $u \equiv ax + by + a\sqrt[3]{b} = 0$ $v \equiv bx - ay + b\sqrt[3]{a} = 0$ $a, b \in \mathbb{R}$ be two straight lines. The equation of the bisectors of the angle formed by $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ for non zero real k_1 and k_2 are:

(A) $u = 0$ (B) $k_2u + k_1v = 0$ (C) $k_2u - k_1v = 0$ (D) $v = 0$

Ans. A, D

Sol. Note that the lines are perpendicular.

Assume the co-ordinate axes to be directed along $u = 0$ and $v = 0$.

Now the lines $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ are equally inclined with u v axes.

Hence the bisectors are $u = 0$ and $v = 0$. i.e. $\frac{k_1u - k_2v}{\sqrt{k_1^2 + k_2^2}} = \pm \frac{k_1u + k_2v}{\sqrt{k_1^2 + k_2^2}}$

Q29. रेखाओं $y - b = \frac{2m}{1 - m^2}(x - a)$ तथा $y - b = \frac{2m'}{1 - m'^2}(x - a)$ के कोणों की अर्द्धक रेखा होगी।

(A) $(y - b)(m + m') + (x - a)(1 - mm') = 0$

(B) $(y - b)(m + m') - (x - a)(1 - mm') = 0$

(C) $(y - b)(1 - mm') + (x - a)(m + m') = 0$

(D) $(y - b)(1 - mm') - (x - a)(m + m') = 0$

Ans. A, D

Sol. Put $m = \tan \alpha$ and $m' = \tan \beta$

Q30. If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s)?

(A) centroid (B) incentre (C) circumcentre (D) orthocentre

Ans. A, C, D

Sol. -